

Cloudflare Radar 2024

Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network in 2024.

Source: Cloudflare Radar | radar.cloudflare.com/year-in-review/2024

17% Global Traffic Growth and Starlink's 3.3x Surge

17%

YoY Traffic Growth

3.3x

Starlink Traffic Surge

41%

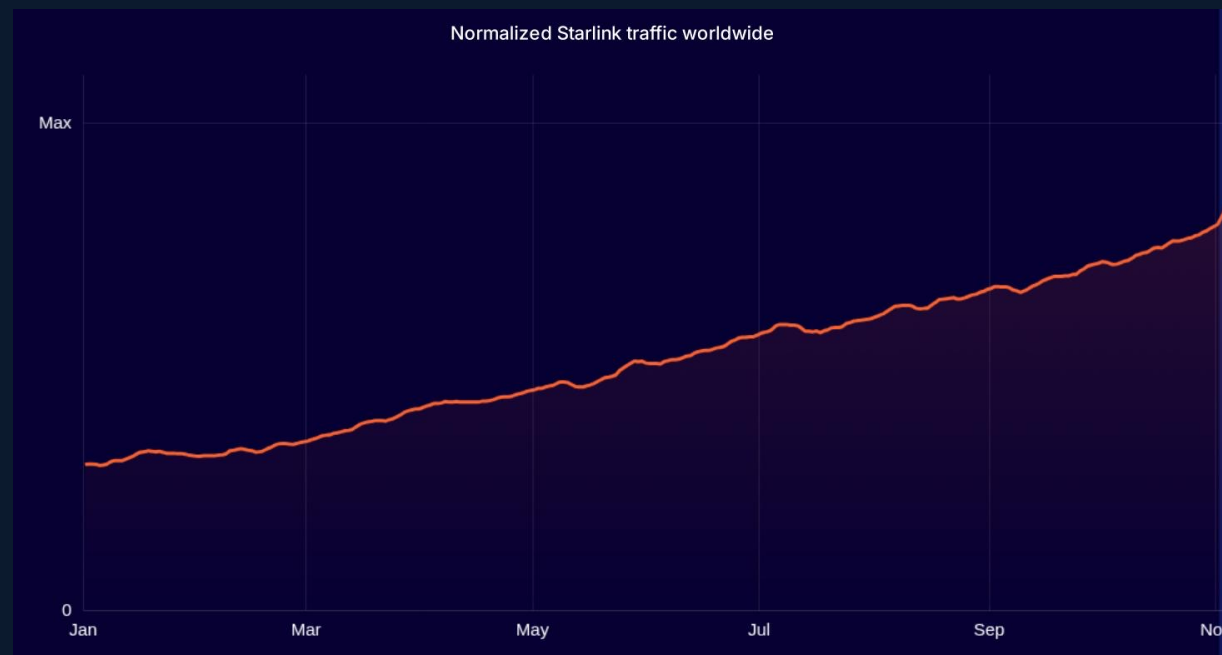
Mobile Traffic Share

28%

IPv6 Adoption

Global Internet traffic grew 17% in 2024, continuing the upward trend driven by increasing reliance on the Internet for communication, commerce, entertainment, and transportation.

SpaceX Starlink traffic saw dramatic 3.3x growth during 2024, continuing to bring satellite Internet connectivity to previously underserved locations.



Google Leads Top 10 Services as Generative AI Rises

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Generative AI

Generative AI continued its rapid rise throughout 2024, while the Metaverse receded into the background. AI-powered services are reshaping the Internet service landscape.

Search Engine Share

Google	89.0%
Yandex	3.1%
Baidu	2.7%
Bing	2.6%
DuckDuckGo	0.95%
Other	2.2%

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Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Post-Quantum Encryption Adoption

Starting at ~2% of TLS 1.3 traffic in early 2024, adoption jumped sharply after Google Chrome 124 enabled post-quantum key agreement by default on April 17, rapidly climbing to ~13%. Mozilla Firefox has also started rolling out post-quantum by default, and Apple Safari began initial testing.

21%

HTTP/3 Adoption

QUIC-based, encrypted by default

33%

iOS Traffic Share

Among mobile devices



Top API Client Languages

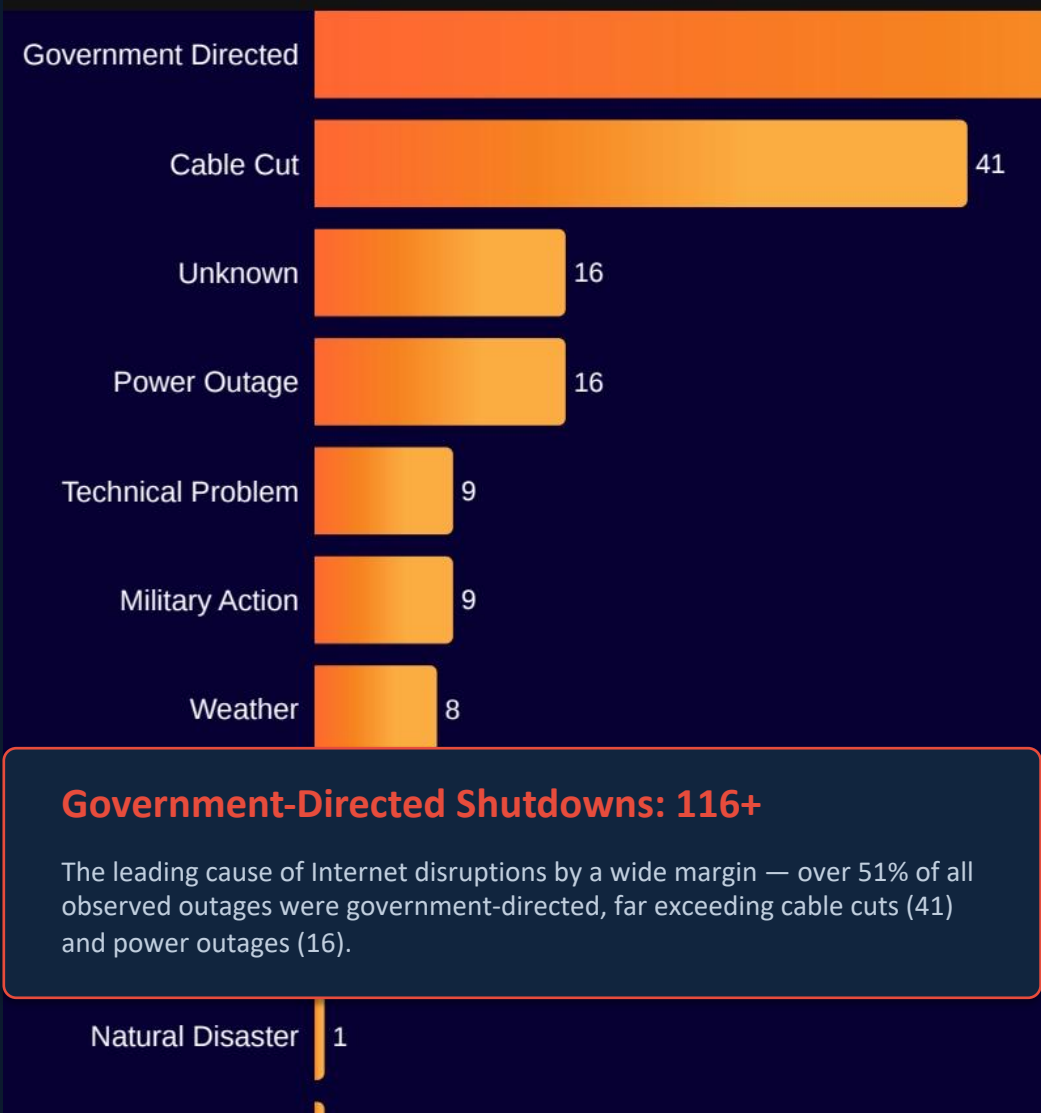
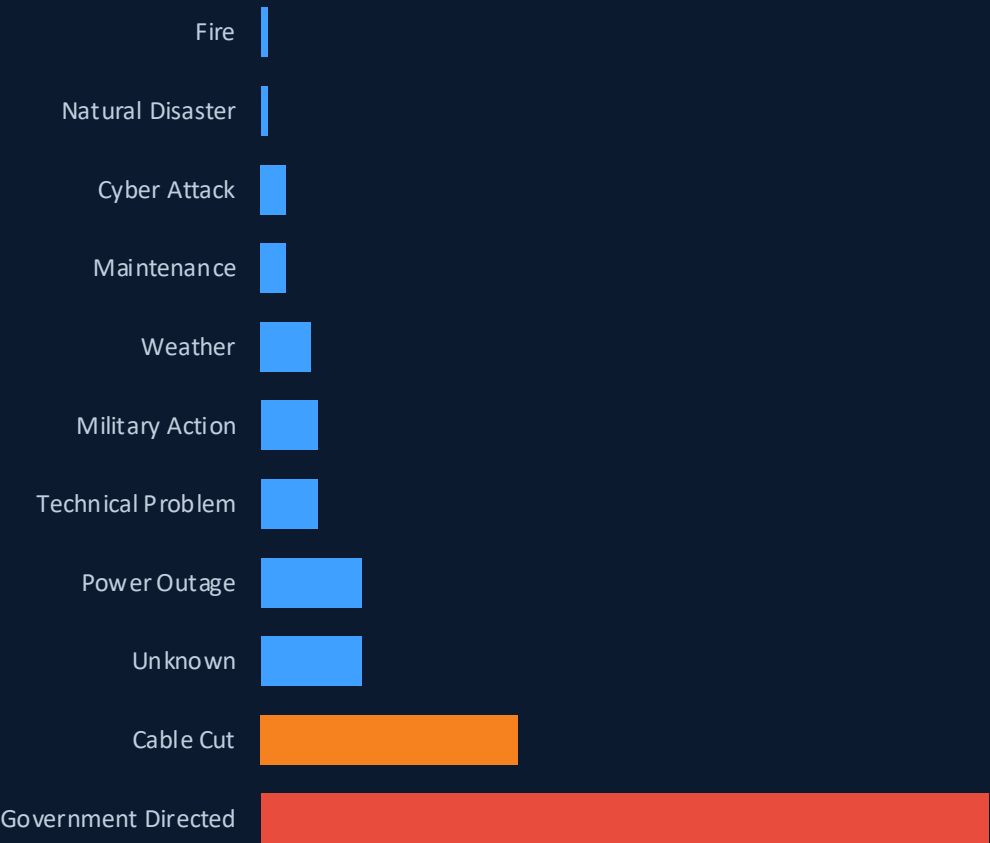
Share %



225 Major Disruptions: Government Shutdowns Lead Outage Causes

Cloudflare observed 225 major Internet disruptions globally in 2024.

Disruptions



6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

Traffic Mitigated

4.3%

Malicious Emails

68%

Bot Traffic from
Top 10 Countries

6.4%

RPKI Route
Growth

Most Attacked Sector: Gambling & Games

The Gambling and Games sector was the most attacked industry category in 2024, reflecting the ongoing targeting of high-transaction, user-heavy platforms by threat actors.

Email Threat Landscape

4.3% of emails processed were malicious

43% of malicious emails contained a deceptive link
Top-level domain .bar originated the largest share of malicious and spam email

Log4j Exploitation Persists (CVE-2021-44228)

First disclosed in December 2021, the Log4j vulnerability continues to be actively targeted by attackers more than two years later, underscoring the persistent nature of widely known vulnerabilities.

Routing Security Improvement

RPKI (Resource Public Key Infrastructure) valid routes increased by 6.4%, an important step toward improved Internet routing security. This growth reflects the industry's continued commitment to securing the BGP routing infrastructure.

Five Takeaways Shaping the 2025 Internet Outlook

1

Traffic Growth Trajectory

Global Internet traffic grew 17% YoY, with Starlink traffic surging 3.3x as satellite connectivity expanded to underserved regions.

2

Post-Quantum Encryption Momentum

Post-quantum encryption adoption leapt from ~2% to 13% of TLS 1.3 traffic following Chrome 124's default enablement in April 2024.

3

Connectivity Fragility from Government Actions

Government-directed shutdowns remained the leading cause of the 225 major Internet disruptions observed, followed by cable cuts (41) and power outages (16).

4

Persistent Security Threats

6.5% of all traffic was mitigated, Gambling/Games was the most targeted sector, and Log4j exploitation persists years after disclosure.

5

Evolving Developer Ecosystem

Go surpassed other languages to become the most popular API client language, capturing roughly 12% of automated API traffic.